HW8

1. (a) $X(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt} dt$

$= \int_0^4 \frac{1}{4} e^{-jwt} dt$

$= \frac{1}{4} \left. \frac{e^{-jwt}}{-jw} \right|_0^4$

$= \frac{1}{4} \left(\frac{e^{-jw4}}{-jw} + \frac{1}{jw} \right)$

$= \frac{1 - e^{-4jw}}{4jw}$

(b) $x(jw) = \int_{-1}^{1} (1 + \cos\pi t) e^{-jwt} dt$

$= \int_{-1}^{1} e^{-jwt} dt + \int_{-1}^{1} \frac{1}{2} e^{(-jw+j\pi)t} dt + \int_{-1}^{1} \frac{1}{2} e^{-(jw+j\pi)t} dt$

$= \left. \frac{e^{-jwt}}{-jw} \right|_{-1}^{1} + \frac{1}{2} \left. \frac{e^{-jw+j\pi}}{-jw+j\pi} \right|_{-1}^{1} + \frac{1}{2} \frac{e^{-jw-j\pi}}{-jw-j\pi}$

$= \frac{2\sin w}{w} + \frac{\sin(w-\pi)}{w-\pi} + \frac{\sin(w+\pi)}{w+\pi}$

(c) $x(jw) = \int_0^{\infty} (1-t^2) e^{-jwt} dt$

$= \left[(1-t^2) \frac{e^{-jwt}}{-jw} + 2t \frac{e^{-jwt}}{-w^2} - 2 \frac{e^{-jwt}}{jw^3} \right]_0^{\infty}$

$\lim_{t \to \infty} e^{-jwt}$ isn't determined

(d) $x(jw) = \int_{-\infty}^{\infty} \sum_{k=0}^{\infty} a^k \delta(t - kT) e^{-jwt} dt$

$= \sum_{k=0}^{\infty} a^k e^{-jwkT}$

$= \frac{1}{1 - \alpha e^{-jwT}}$

(e) $x(jw) = \int_{-1}^{0} (1+t) e^{-jwt} dt + \int_0^1 (1-t) e^{-jwt}$

$= \left[(1+t) \frac{e^{-jwt}}{-jw} + \frac{e^{-jwt}}{w^2} \right]_{-1}^{0} + \left[(1-t) \frac{e^{-jwt}}{-jw} - \frac{e^{-jwt}}{w^2} \right]_0^1$

$= \frac{1}{-jw} + \frac{1}{w^2} + \quad - \frac{e^{+jw}}{w^2} - \frac{e^{-jw}}{w^2} + \frac{1}{jw} + \frac{1}{w^2}$

$= \frac{2 - 2\cos wt}{w^2}$

(f) $x(t) = \sum_{n=-\infty}^{\infty} \delta(t-n) + \sum_{n=-\infty}^{\infty} \delta(t-2m)$

$x(jw) = 2\pi \sum_{n=-\infty}^{\infty} \delta(w - 2\pi k) + \pi \sum_{n=-\infty}^{\infty} \delta(w - \pi k)$

2. $x(jw) = \frac{2\sin(3(w-2\pi))}{w-2\pi}$

$x(j(w-2\pi)) = \frac{2\sin 3w}{w}$

according to P1.(a), the structure are

quite same $x(t) = u(t+3) - u(t-3)$

$x(jw_0) = \frac{e^{-jw3} - e^{+jw3}}{-jw} = \frac{\sin 3w}{w}$

$\therefore x(t) = e^{j2\pi t} [u(t+3) - u(t-3)]$

Heyang Zhang bracket_x@163.com